

Number System

1. Divisibility Rules

Number Rule

2	Last digit divisible by 2 (0,2,4,6,8)
3	Sum of digits divisible by 3
4	Last 2 digits divisible by 4
5	Last digit = 0 or 5
6	Divisible by both 2 and 3
8	Last 3 digits divisible by 8
9	Sum of digits divisible by 9
10	Last digit = 0
11	$ (\text{Sum of odd place digits}) - (\text{Sum of even place digits}) = 0$ or multiple of 11
7	Remove last digit d and compute Remaining + 5d repeatedly
13	Remove last digit d and compute Remaining + 4d repeatedly
17	Remove last digit d and compute Remaining – 5d repeatedly
19	Remove last digit d and compute Remaining + 2d repeatedly

2. HCF & LCM

For Two Numbers

Product Formula

$$HCF \times LCM = a \times b$$

Finding One When Other is Known

$$LCM = \frac{a \times b}{HCF}$$
$$HCF = \frac{a \times b}{LCM}$$

For Fractions

For fractions: $\frac{a}{b}, \frac{c}{d}$

$$HCF = \frac{HCF(a, c)}{LCM(b, d)}$$

$$LCM = \frac{LCM(a, c)}{HCF(b, d)}$$

3. Prime Factorization & Factors

If $N = p^a q^b r^c$ where p,q,r are prime numbers.

Total Number of Factors

$$d(N) = (a + 1)(b + 1)(c + 1)$$

Sum of Factors

$$\sigma(N) = \frac{p^{a+1} - 1}{p - 1} \times \frac{q^{b+1} - 1}{q - 1} \times \frac{r^{c+1} - 1}{r - 1}$$

Product of Factors $P = N^{d(N)/2}$

Where $d(N)$ = total number of factors

Number of Prime Factors

Distinct primes only: $N = 2^3 \times 3^2 \times 5$

Prime factors = 3

Odd Factors

Ignore power of 2.

$$(a_{odd} + 1)(b_{odd} + 1) \cdots$$

Even Factors

$$\text{Even Factors} = \text{Total Factors} - \text{Odd Factors}$$

4. Factorials

Definition $N! = N(N - 1)(N - 2) \cdots 2 \cdot 1$

$$0! = 1, 1! = 1$$

Trailing Zeros in N!

$$Z = \left\lfloor \frac{N}{5} \right\rfloor + \left\lfloor \frac{N}{25} \right\rfloor + \left\lfloor \frac{N}{125} \right\rfloor + \cdots$$

Highest Power of Prime p in N!

(Legendre's Formula)

$$v_p(N!) = \left\lfloor \frac{N}{p} \right\rfloor + \left\lfloor \frac{N}{p^2} \right\rfloor + \left\lfloor \frac{N}{p^3} \right\rfloor + \cdots$$

5. Remainder & Divisibility Theorems

Basic Results $a^n - b^n$ is always divisible by $a - b$

Divisibility by (a+b)

$a^n + b^n$ is divisible by $a + b$

only when n is odd.

$a^n - b^n$ is divisible by $a + b$

only when n is even.

Fermat's Little Theorem

If p is prime and $\gcd(a, p) = 1$

$$a^{p-1} \equiv 1 \pmod{p}$$

Wilson's Theorem

If p is prime $(p - 1)! \equiv -1(\text{mod } p)$ or $(p - 1)! \equiv p - 1(\text{mod } p)$

6. Unit Digit Formulas

Product Unit Digit

Only multiply unit digits.

Example: $37 \times 48 \times 29$

Unit digit: $7 \times 8 \times 9 = 504$

Answer = 4

Cyclicity Method

Cycle Length = 1

Last Digit Unit Digit of Any Power

0	0
1	1
5	5
6	6

Cycle Length = 2

Digit Pattern

4	4,6
9	9,1

Use:

$$n \text{ mod } 2$$

Cycle Length = 4

Digit Pattern

2	2,4,8,6
3	3,9,7,1

Digit Pattern

7 7,9,3,1

8 8,4,2,6

Use:

$$n \bmod 4$$

If remainder = 0, use 4.

7. Number of Digits

Digits in a Number N

$$\text{Digits} = \lfloor \log_{10} N \rfloor + 1$$

Digits in N^k

$$\text{Digits} = \lfloor k \log_{10} N \rfloor + 1$$

8. Base Conversion Formulas

Decimal to Any Base b

Repeated division by b.

Any Base to Decimal

For

$$(a_n a_{n-1} \dots a_0)_b$$
$$N = a_n b^n + a_{n-1} b^{n-1} + \dots + a_0$$

9. Important Properties

Consecutive Numbers

$$HCF(n, n+1) = 1$$

Consecutive Even Numbers

$$HCF(2n, 2n + 2) = 2$$

Product of Consecutive Numbers

$n(n + 1)$ is always even.

Product of Three Consecutive Numbers

$n(n + 1)(n + 2)$ is always divisible by 6.

Product of Four Consecutive Numbers

$n(n + 1)(n + 2)(n + 3)$ is always divisible by 24.

Percentage, Profit and Loss

1. Percentages: The Foundation

The golden rule for percentage problems in aptitude tests: **The base taken is always the original quantity. Core Conversions:**

- **Fraction to Percentage:** Multiply by 100.
- **Percentage to Fraction/Decimal:** Divide by 100.

High-Frequency Placement Patterns:

- **Successive Percentage Change:** If a value changes by $a\%$ and then by $b\%$, the net percentage change is:

$$NetChange = a + b + \frac{ab}{100}$$

(Note: Use negative signs for decreases or losses).

- **Product Constancy:** If price increases by $x\%$, consumption must decrease by $\frac{x}{100+x} \times 100\%$ to keep expenditure constant.

2. Profit, Loss, and Discount

Most placement questions here will try to trick you by mixing Cost Price (CP), Selling Price (SP), and Marked Price (MP).

Core Formulas:

- **Profit / Loss:** Always calculated on **Cost Price (CP)** unless stated otherwise.

$$\text{Profit \%} = \frac{SP - CP}{CP} \times 100$$

$$\text{Loss \%} = \frac{CP - SP}{CP} \times 100$$

- **Direct SP/CP Calculation:**

$$SP = CP \times \left(\frac{100 \pm \text{Profit/Loss \%}}{100} \right)$$

$$CP = SP \times \left(\frac{100}{100 \pm \text{Profit/Loss \%}} \right)$$

- **Discounts & Markups:** * Discount is always given on **Marked Price (MP)**.

$$\text{Discount \%} = \frac{MP - SP}{MP} \times 100$$

- Markup is always calculated on **Cost Price (CP)**.

$$\text{Markup \%} = \frac{MP - CP}{CP} \times 100$$

High-Frequency Placement Pattern:

- **The MP/CP Master Relation:** When a question gives you both the discount percentage and the profit percentage, skip the long math and use this direct ratio:

$$\frac{MP}{CP} = \frac{100 + \text{Profit \%}}{100 - \text{Discount \%}}$$

3. Simple & Compound Interest (SI & CI)

Aptitude rounds love testing the difference between SI and CI because it checks your ability to handle compounding math quickly.

Core Formulas:

- **Simple Interest (SI):**

$$SI = \frac{P \times R \times T}{100}$$

- **Compound Interest (CI):**

$$\text{Amount (A)} = P \left(1 + \frac{R}{100} \right)^n$$

$$CI = P \left(1 + \frac{R}{100} \right)^n - P$$

High-Frequency Placement Patterns:

- **Difference between CI and SI for 2 Years:** This is one of the most common direct questions in online assessments.

$$\text{Difference (D)} = P \left(\frac{R}{100} \right)^2$$

- **Difference between CI and SI for 3 Years:**

$$\text{Difference (D)} = P \left(\frac{R}{100} \right)^2 \left(3 + \frac{R}{100} \right)$$

- **Rule of 72 (Approximation):** To find out how many years it takes for your money to double at a given compound interest rate R , divide 72 by R (e.g., at 8% interest, it takes roughly 9 years).

Ratio and proportion

1. The Proportion Core

Almost every missing-value question boils down to the fundamental rule of proportions:

Product of Extremes = Product of Means.

If $a:b::c:d$, then:

$$a \times d = b \times c$$

High-Frequency Direct Calculations:

Expect direct questions asking for these specific terms to test your vocabulary and speed:

- **Mean Proportional** (between a and b): Let it be x .

$$x = \sqrt{ab}$$

- **Third Proportional** (to a and b): Let it be c .

$$c = \frac{b^2}{a}$$

- **Fourth Proportional** (to a, b , and c): Let it be d .

$$d = \frac{bc}{a}$$

2. The "Work & Efficiency" Flip

This is a favorite for companies testing logical speed. The rule is simple: **Efficiency is inversely proportional to time.**

- **Pattern:** If A is twice as fast as B, their efficiency ratio is 2: 1. Therefore, the ratio of the time they take to complete the same work is exactly flipped: 1: 2.
- **Formula Application:** If the ratio of efficiencies of A and B is $x: y$, the ratio of days taken to complete the job is:

$$\text{Time Ratio} = y: x$$

3. Partnerships: The Ultimate Ratio Application

Partnership questions are guaranteed to appear. They look like massive word problems but are solved with one single logic matrix: **Profit depends on how much you invested, and how long you left it there.**

Simple Partnership (Same Time Period):

When all partners invest for the same duration, profits are split purely based on the money invested.

- Profit Ratio = Investment_A: Investment_B

Compound Partnership (Different Time Periods):

This is the pattern you will see 90% of the time. You must multiply the capital by the time it was active.

- **The Master Formula:** If A invests amount I_1 for T_1 months, and B invests amount I_2 for T_2 months:

$$\frac{\text{Profit}_A}{\text{Profit}_B} = \frac{I_1 \times T_1}{I_2 \times T_2}$$

- *Trap Warning:* Watch out for phrasing like "A joins 3 months later" (meaning A's money was only active for 9 months out of a year) or "B withdraws half his capital after 6 months" (meaning you have to split B's calculation into two phases and add them).

4. Advanced "Merge" Ratios (Compound Ratios)

When tests want to combine multiple conditions, they use compound ratios.

- **Compound Ratio** of $a:b$ and $c:d$ is simply the product of their antecedents (first terms) and consequents (second terms):

$$\text{Result} = ac:bd$$

- **Pattern Shortcut:** If $A:B = 2:3$ and $B:C = 4:5$, don't calculate them separately. Find a common multiple for B (in this case, 12). Multiply the first ratio by 4 and the second by 3 to get $A:B:C = 8:12:15$.

Probability and Combinatorics

1. The "AND/OR" Golden Rule

Before applying any complex formula, remember the fundamental principle of counting. This solves half of the basic questions.

- **"AND" means Multiply:** If you need to do Task A *and* Task B, you multiply their possibilities ($m \times n$).
- **"OR" means Add:** If you need to do Task A *or* Task B, you add their possibilities ($m + n$).

2. Permutation & Combination (P&C) Patterns

Pattern 1: Arranging Words with Repeating Letters

This is arguably the most common P&C question (e.g., "In how many ways can the letters of the word MISSISSIPPI be arranged?").

- **Formula:** Divide the total factorial by the factorials of the repeating items.

$$\text{Arrangements} = \frac{n!}{p! \cdot q! \cdot r!}$$

Pattern 2: Circular Arrangements

Aptitude tests love putting people around tables.

- **People around a table:** Because there is no fixed starting point, the formula shifts.

$$\text{Ways} = (n - 1)!$$

- **Beads on a necklace / Flowers in a garland:** Clockwise and anti-clockwise arrangements are identical, so you divide by 2.

$$\text{Ways} = \frac{(n - 1)!}{2}$$

Pattern 3: The Handshake / Tournament Problem

"If there are n people in a room and everyone shakes hands with everyone else exactly once, how many handshakes occur?"

- **Formula:** This is a basic combination problem of selecting 2 people from n .

$${}^nC_2 = \frac{n(n - 1)}{2}$$

- **Geometry Application:** This same formula gives you the number of straight lines you can draw connecting n non-collinear points. For diagonals in an n -sided polygon, use:

$$\text{Diagonals} = \frac{n(n - 3)}{2}$$

Pattern 4: Total Possible Selections

When a question asks for the number of ways to select *at least one* item from n distinct items.

- **Formula:**

$${}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n - 1$$

3. Probability Patterns

Probability is essentially combinations applied to fractions. The core logic is always:

$$P(E) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Possible Outcomes}}$$

High-Frequency Data Sets to Memorize

Item	Key Facts for Quick Recall
Coins	Total outcomes for n coins tossed = 2^n .
Dice	Total outcomes for n dice rolled = 6^n . Sum of 7 is the most likely outcome when rolling two dice (occurs 6 times).
Cards	52 total cards. 4 suits (Spades, Clubs, Hearts, Diamonds) of 13 cards each. 12 Face Cards total (Kings, Queens, Jacks).

Pattern 1: The "At Least One" Trick

If a question asks for the probability of getting *at least one* specific outcome (e.g., "probability of getting at least one head in 4 tosses"), do not calculate 1 head, 2 heads, 3 heads, and 4 heads. Use the complementary event.

- **Formula:**

$$P(\text{At least one}) = 1 - P(\text{None})$$

Pattern 2: Drawing Items with vs. without Replacement

Most "bag of colored balls" questions test this.

- **Without Replacement (Dependent Events):** The denominator decreases with each draw. If drawing 2 red balls from a bag of 5 red and 7 blue, the probability is $\frac{5}{12} \times \frac{4}{11}$.
- **With Replacement (Independent Events):** The denominator stays the same. The probability is $\frac{5}{12} \times \frac{5}{12}$.

Pattern 3: The Addition Theorem (Mutually Exclusive vs. Not)

- **Mutually Exclusive:** Events cannot happen at the same time (e.g., drawing a card that is both a King and a Queen).

$$P(A \cup B) = P(A) + P(B)$$

- **Not Mutually Exclusive:** Events have overlap (e.g., drawing a card that is a King or a Heart — there is one King of Hearts). You must subtract the overlap.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Time and work

1. The LCM Method (The Golden Rule)

Instead of dealing with fractions, assume the **Total Work** is a concrete number: the LCM of all the days given in the problem.

- **The Core Equation:**

$$\text{Total Work} = \text{Efficiency} \times \text{Time}$$

- **Efficiency:** This is the amount of work done in *one day*. If A takes 10 days and B takes 15 days, Total Work = 30 units (LCM of 10 and 15). A's efficiency is 3 units/day, B's is 2 units/day.

2. High-Frequency Direct Formulas

If you want to skip the LCM setup for straightforward questions, memorize these direct calculation shortcuts:

- **Two People Working Together:** If A takes x days and B takes y days, the time taken together is:

$$\text{Time} = \frac{xy}{x + y}$$

- **Three People Working Together:** If A, B, and C take x , y , and z days respectively:

$$\text{Time} = \frac{xyz}{xy + yz + zx}$$

- **The "MDH" Rule (Men-Days-Hours):** Used when comparing two different groups doing a piece of work.

$$\frac{M_1 \times D_1 \times H_1}{W_1} = \frac{M_2 \times D_2 \times H_2}{W_2}$$

(Where M = Men/Workers, D = Days, H = Hours/Day, and W = Amount of Work).

3. Most Frequently Tested Patterns

Pattern 1: The "Leaving/Joining Mid-Way" Problem

- **The Setup:** A and B start working together, but A leaves after n days.
- **The Shortcut:** Calculate the work completed in those n days by both, subtract it from the Total Work (LCM), and divide the remaining work by the efficiency of the person left behind (B).

Pattern 2: Efficiency Ratios

- **The Setup:** "A is 3 times as efficient as B" or "A is 50% more efficient than B".
- **The Rule:** Efficiency and Time are strictly inversely proportional. If the efficiency ratio of A to B is 3:1, the ratio of the time they take to finish the same work is exactly flipped: 1:3.

Pattern 3: Alternate Days Working

- **The Setup:** A and B work on alternate days (A on day 1, B on day 2, etc.).
- **The Rule:** Find their combined work for a **2-day cycle**. Divide the Total Work by this cycle's work to find the number of full cycles, then calculate the remaining units manually for the final days.

4. Pipes and Cisterns: The Evil Twin

This is the exact same concept as Time and Work, with one critical twist: **Leaks or emptying pipes have a negative efficiency.**

- **Pattern:** Pipe A fills a tank in x hours, but due to a leak at the bottom (Pipe B), it takes y hours.
- **Formula:** The time taken by the leak to empty a full tank alone is:

$$\text{Time} = \frac{xy}{y - x}$$

Time, Speed and Distance

1. The Core Fundamentals & Conversions

Everything in this topic stems from one universal equation:

$$D = S \times T$$

Crucial Unit Conversions (Do not skip this step in exams):

- **km/hr to m/s:** Multiply by $\frac{5}{18}$
- **m/s to km/hr:** Multiply by $\frac{18}{5}$

(Memory trick: $18 \text{ km/hr} = 5 \text{ m/s}$. Since m/s is a smaller unit, the fraction with the smaller numerator ($\frac{5}{18}$) is used to convert down).

2. Average Speed Patterns

Aptitude tests almost never ask for a simple average. They will give you split journeys to test if you know that Average Speed is NOT the average of the speeds.

- **Universal Formula:**

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

- **Pattern 1: Equal Distances, Two Speeds:** (e.g., Going from home to office at x km/hr and returning at y km/hr).

$$\text{Average Speed} = \frac{2xy}{x + y}$$

- **Pattern 2: Equal Distances, Three Speeds:** (e.g., Covering three equal parts of a journey at speeds x , y , and z).

$$\text{Average Speed} = \frac{3xyz}{xy + yz + zx}$$

3. Relative Speed

When two objects are moving, you must merge their speeds into a single "relative" speed.

- **Moving in the Same Direction:** (e.g., Police chasing a thief).

$$\text{Relative Speed} = S_1 - S_2 \quad (\text{where } S_1 > S_2)$$

- **Moving in Opposite Directions:** (e.g., Two cars driving towards each other).

$$\text{Relative Speed} = S_1 + S_2$$

4. The "Early / Late" Distance Shortcut

The Setup: "If a man travels at S_1 he reaches 10 minutes late, but if he travels at S_2 he reaches 5 minutes early. Find the distance."

This is one of the most frequently asked questions. Do not solve it with linear equations; use this direct formula:

$$D = \frac{S_1 \times S_2}{|S_1 - S_2|} \times \Delta T$$

(Note: ΔT is the total time difference. If one is early and one is late, add the times. Ensure ΔT is converted to hours if speeds are in km/hr).

5. Trains Crossing Objects

Train problems are just TSD problems where the **length of the train matters** as part of the distance. Let L_1 be the length of the first train and L_2 be the length of the second object.

- **Pattern 1: Crossing a stationary point** (pole, standing man, signal post).
 - Distance = L_1
- **Pattern 2: Crossing a stationary length** (platform, bridge, tunnel).
 - Distance = $L_1 + L_2$
- **Pattern 3: Two trains crossing each other.**
 - Distance is ALWAYS = $L_1 + L_2$ (regardless of direction).
 - Apply Relative Speed rules for the denominator (Add if opposite, subtract if same direction).

$$T = \frac{L_1 + L_2}{S_1 \pm S_2}$$

6. Boats and Streams

This flips the logic of Relative Speed. Instead of two objects moving, one object is moving *on top* of a moving surface.

Let B = Speed of boat in still water, and S = Speed of the stream/river.

- **Downstream (With the flow):** Speed (D) = $B + S$
- **Upstream (Against the flow):** Speed (U) = $B - S$

High-Frequency Direct Formulas:

If the question gives you the Downstream (D) and Upstream (U) speeds directly, you can instantly find the boat or stream speed:

$$B = \frac{D + U}{2}$$

$$S = \frac{D - U}{2}$$

7. Advanced Placement Hacks

Pattern 1: The "Cross-Over" After Meeting

The Setup: Two trains start at the same time from points A and B towards each other. After crossing each other, they take T_1 and T_2 hours to reach their respective destinations.

$$\frac{S_1}{S_2} = \sqrt{\frac{T_2}{T_1}}$$

Pattern 2: Stoppage Time

The Setup: "Excluding stoppages, a bus travels at 60 km/hr, and including stoppages, it travels at 45 km/hr. How many minutes does it stop per hour?"

$$\text{Stoppage Time (per hour)} = \frac{\text{Fast Speed} - \text{Slow Speed}}{\text{Fast Speed}} \times 60$$

Pattern 3: Circular Tracks (First Meeting Time)

If two runners start from the same point on a circular track of circumference C at speeds S_1 and S_2 :

$$\text{Time to meet for the first time} = \frac{C}{\text{Relative Speed}}$$

Progressions (AP, GP..)

1. Arithmetic Progression (AP)

An AP is a sequence where the difference between consecutive terms is constant (d).

Let a be the first term, d be the common difference, and l be the last term.

- **n -th Term:**

$$T_n = a + (n - 1)d$$

- **Sum of n Terms (Standard):**

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

- **Sum of n Terms (If last term is known):**

$$S_n = \frac{n}{2}(a + l)$$

- **Arithmetic Mean (AM) of two numbers a and b :**

$$AM = \frac{a + b}{2}$$

High-Frequency AP Patterns:

- **The "Zero Sum" Trick:** If the sum of the first p terms is equal to the sum of the first q terms, then the sum of the first $(p + q)$ terms is exactly 0.
- **The "Swap" Trick:** If the p -th term is q , and the q -th term is p , then the $(p + q)$ -th term is exactly 0.
- **Assumption Shortcut:** If a question requires you to assume 3 numbers in an AP, always use $(a - d), a, (a + d)$. For 4 numbers, use $(a - 3d), (a - d), (a + d), (a + 3d)$. This makes the d cancel out when you sum them.

2. Geometric Progression (GP)

A GP is a sequence where the ratio of consecutive terms is constant (r).

- **n -th Term:**

$$T_n = ar^{n-1}$$

- **Sum of n Terms (If $r > 1$):**

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

- **Sum of n Terms (If $r < 1$):**

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

- **Geometric Mean (GM) of two numbers a and b :**

$$GM = \sqrt{ab}$$

High-Frequency GP Pattern:

- **Infinite Geometric Series:** This is one of the most frequently tested concepts, often disguised as a bouncing ball problem (e.g., "A ball is dropped from 100m and bounces back to 80% of its height... find total distance traveled").

If $|r| < 1$, the sum to infinity is:

$$S_{\infty} = \frac{a}{1 - r}$$

3. Harmonic Progression (HP)

A sequence is in HP strictly if the *reciprocals* of its terms form an AP. There is no direct formula for the sum of an HP; you must convert it to an AP, solve, and invert if necessary.

- **Standard Rule:** If a, b, c are in HP, then $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP.
- **Harmonic Mean (HM) of two numbers a and b :**

$$HM = \frac{2ab}{a + b}$$

(Note: You will recognize this exact formula from the Time, Speed, and Distance section for finding Average Speed over equal distances!)

4. The AM-GM-HM Master Relationship

When companies want to test higher-level quantitative logic, they use inequalities based on the means of positive real numbers.

- **The Order Inequality:** For any set of positive numbers (unless they are all identical):

$$AM > GM > HM$$

- **The Multiplicative Identity:** For any two positive numbers:

$$AM \times HM = GM^2$$

- **Minimum Value Trick:** The minimum value of $x + \frac{1}{x}$ (where $x > 0$) is always 2, which is derived directly from $AM \geq GM$.

5. Special Series (Must-Memorize Sums)

Tests frequently ask for the sum of standard sequences. Do not manually calculate these; commit them to memory.

- **Sum of the first n natural numbers:**

$$\Sigma n = \frac{n(n + 1)}{2}$$

- **Sum of the squares of the first n natural numbers:**

$$\Sigma n^2 = \frac{n(n + 1)(2n + 1)}{6}$$

- **Sum of the cubes of the first n natural numbers:**

$$\Sigma n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

- **Sum of the first n odd numbers:**

$$S_n = n^2$$

- **Sum of the first n even numbers:**

$$S_n = n(n+1)$$

Ages

1. The Core Rules of Ages

Before touching a formula, you must internalize these three unbreakable rules:

- **The Constant Difference:** The difference in age between two people **never changes**, regardless of whether you are in the past, present, or future. (If you are 5 years older than your brother today, you were 5 years older ten years ago, and will be 5 years older in fifty years).
- **The "Present" Anchor:** Always let the **present age** of the unknown person be x . Never set x as the past or future age, or you will get tangled in negative signs.
- **The Group Shift:** If the sum of the ages of n people is S today, the sum of their ages after y years is not $S + y$; it is $S + (n \times y)$.

2. The Ultimate Ratio Shortcut (The Cross-Product Method)

This is the most frequently tested pattern in placement exams.

The Setup: "The present ratio of ages of A and B is $a:b$. After t years, the ratio becomes $c:d$. Find their present ages."

Instead of solving the algebraic equation $\frac{ax+t}{bx+t} = \frac{c}{d}$, use this direct shortcut to find the multiplier x :

$$x = \frac{t \times (c - d)}{\text{Difference of cross products } (ad - bc)}$$

(Note: Always take the absolute, positive difference for both numerator and denominator).

Once you have x , A's present age is ax and B's present age is bx .

(This exact formula also works for "t years ago" - just use t as the number of years).

3. High-Frequency Placement Patterns

Pattern 1: The "Times/Multiple" Shift

The Setup: "A is 3 times as old as B. After 10 years, A will be twice as old as B."

The Standard Setup:

- Present: $A = 3x$, $B = x$
- Future: $(3x + 10) = 2(x + 10)$

(Solve for x : $3x + 10 = 2x + 20 \Rightarrow x = 10$. So B is 10, A is 30).

Pattern 2: Past Ratio to Future Ratio (The Double Shift)

The Setup: "5 years ago, the ratio of A to B was 2:3. In 5 years from now, the ratio will be 3:4."

The Trick: The total time gap t between the two ratios is **10 years** (5 past + 5 future).

You can use the Ultimate Ratio Shortcut from above, setting $t = 10$.

$$x = \frac{10 \times (4 - 3)}{(3 \times 3) - (2 \times 4)} = \frac{10 \times 1}{9 - 8} = 10$$

Careful: This x belongs to the *past* ratio. So 5 years ago, A was $2(10) = 20$. Present age of A = $20 + 5 = 25$.

Pattern 3: The "As much older as he is younger" Trap

The Setup: "A is as much older than B as he is younger than C."

The Rule: This phrasing means A is exactly in the middle of B and C. Therefore, A's age is the arithmetic mean (average) of B and C's ages.

$$\text{Age of A} = \frac{\text{Age of B} + \text{Age of C}}{2}$$

4. Average Age Patterns

Pattern 1: The Aging Group

- If the average age of n people is A , then after y years, the new average age is exactly $A + y$.
- y years ago, the average was $A - y$.

Pattern 2: The Replacement Member

The Setup: "In a family of 5, a 60-year-old is replaced by a new person, and the average age of the family drops by 2 years. What is the age of the new person?"

The Formula:

$$\text{Age of New Person} = \text{Age of Replaced Person} \pm (\text{Change in Average} \times \text{Total Members})$$

(Use $+$ if the average increased, use $-$ if the average decreased).

Example: Age = $60 - (2 \times 5) = 60 - 10 = 50$ years old.